1 The density matrix, entanglement entropy and their connection to singular value decomposition (SVD)

1.1 Main goal

Prove that, for a bipartite pure state, the von Neumann entanglement entropy is

$$\boxed{S = -\operatorname{Tr}\big(\rho_A \log \rho_A\big) = -\sum_i \lambda_i \log \lambda_i = -\sum_i s_i^2 \log s_i^2}, \tag{1}$$

where λ_i are the eigenvalues of the reduced density matrix ρ_A and s_i are the Schmidt values, with $\lambda_i = s_i^2$. We use the convention $0 \log 0 = 0$.

1.2 The density operator

Let $\{p_i, |\psi_i\rangle\}$ be an ensemble of pure states, describing a system that is in one of the states $|\psi_i\rangle$ with probability p_i, i.e. the system is not fully known. Define the density operator as

$$\rho = \sum_i p_i \, |\psi_i\rangle\langle\psi_i| \,, \qquad p_i \in \mathbb{R} \;\; \& \;\; \sum_i p_i = 1, \tag{2}$$

which is a positive semi-definite operator of trace one (see for example Nielsen & Chuang).

1.3 Observables on a composite system

Take a system A with some observable M, and a system B, giving the composite system AB (see Nielsen & Chuang Box 2.6). Assume we can measure M on system A; then a corresponding measurement on AB is $\tilde{M} = M \otimes \mathbb{1}_B$.

Proof. Take AB to be prepared in $|m\rangle\,|\psi\rangle$ with $M\,|m\rangle = m\,|m\rangle$. Applying the measurement to $|m\rangle\,|\psi\rangle$ should yield m with probability one, i.e. the corresponding projector for $\tilde{M}$ is $P_m \otimes \mathbb{1}_B$. By the spectral theorem, $\tilde{M} = \sum_m m\, P_m \otimes \mathbb{1}_B = M \otimes \mathbb{1}_B$ for a general state of AB. □

1.4 The reduced density matrix as a partial trace

Measurement averages must be the same whether computed via the full ρ^{AB} or via some ρ^A associated to system A, i.e.

$$\operatorname{tr}\big(M\rho^A\big) \stackrel{!}{=} \operatorname{tr}\big(\tilde{M}\rho^{AB}\big) = \operatorname{tr}\big((M \otimes \mathbb{1}_B)\rho^{AB}\big). \tag{3}$$

Take $\rho^A = \operatorname{tr}_B(\rho^{AB})$ where

$$\operatorname{tr}_B\big(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|\big) \equiv |a_1\rangle\langle a_2| \, \operatorname{tr}\big(|b_1\rangle\langle b_2|\big) \qquad \& \qquad \operatorname{tr}_B(\cdot) \text{ linear.} \tag{4}$$

Write ρ^{AB} by using the eigenbasis of M for system A with $M|i\rangle = m_i|i\rangle$ and $\langle i|j\rangle = \delta_{ij}$:

$$\rho^{AB} = \sum_\alpha p^\alpha |\psi\rangle\langle\psi|^\alpha = \sum_\alpha p^\alpha \sum_{ijk\ell} c_{ij}^\alpha (c_{k\ell}^\alpha)^* \; |i\rangle_A |j\rangle_B \langle\ell|_B \langle k|_A \,, \tag{5}$$

$$\rho^A = \sum_\alpha \sum_{ijk\ell} p^\alpha c_{ij}^\alpha (c_{k\ell}^\alpha)^* \; |i\rangle\langle k|_A \underbrace{\mathrm{tr}\big(|j\rangle\langle\ell|_B\big)}_{=\delta_{j\ell}} = \sum_{\alpha,ijk} p^\alpha c_{ij}^\alpha (c_{kj}^\alpha)^* \; |i\rangle\langle k|_A \,. \tag{6}$$

Therefore

$$\mathrm{tr}\big(M\rho^A\big) = \sum_{\alpha,ijk} p^\alpha c_{ij}^\alpha (c_{kj}^\alpha)^* \; \underbrace{\mathrm{tr}\big(m_i |i\rangle\langle k|_A\big)}_{=m_i\delta_{ik}} = \sum_{\alpha,ij} p^\alpha m_i \, c_{ij}^\alpha (c_{ij}^\alpha)^*, \tag{7}$$

and similarly $\mathrm{tr}\big((M\otimes \mathbb{1}_B)\rho^{AB}\big) = \sum_{\alpha,ij} p^\alpha m_i \, c_{ij}^\alpha (c_{ij}^\alpha)^*$, which proves the claim. One can also easily show that this is the only map that conserves consistency (see Nielsen & Chuang Box 2.6).

1.5 Reduced density matrix of a pure state

Now examine the reduced density matrix for a pure state,

$$\rho_{AB} = |\psi\rangle\langle\psi| = \sum_{ijk\ell} c_{ij}(c_{k\ell})^* \; |i\rangle_A |j\rangle_B \langle\ell|_B \langle k|_A \,, \tag{8}$$

$$\rho_A = \mathrm{tr}_B(\rho_{AB}) = \sum_{ijk} c_{ij}(c_{kj})^* \; |i\rangle\langle k|_A \,, \tag{9}$$

such that the matrix representation is

$$\boxed{\rho_A = \psi\psi^\dagger}, \tag{10}$$

using that the c_{ij} parametrize the matrix $\psi \in \mathbb{C}^{d_A\times d_B}$, i.e. $|\psi\rangle = \sum_{ij} c_{ij}|i\rangle_A |j\rangle_B$ with $\sum_{ij}|c_{ij}|^2 = 1$.

Furthermore, eq. (9) results in

$$\begin{aligned}
\mathrm{tr}\,\rho_A &= \mathrm{tr}\left[\sum_{ijk} c_{ij}(c_{kj})^* |i\rangle\langle k|_A\right] \\
&= \sum_{ijkl} c_{ij}(c_{kj})^* \delta_{il}\delta_{kl} \\
&= \sum_{jl} c_{lj}(c_{lj})^* = 1,
\end{aligned}$$

which means that that the eigenvalues of the reduced density matrix sum to one. This can be seen using that any hermitian matrix can be diagonalized and the cylic property of the trace

$$\mathrm{tr}\,\rho_A = \mathrm{tr}[ODO^\dagger] = \mathrm{tr}\,D = \sum_i \lambda_i = 1. \tag{11}$$

1.6 Connection to the SVD

Now use the SVD on ψ:

$$\psi = USV^\dagger \qquad \text{with } U \in \mathbb{C}^{d_A \times d_A},\ V \in \mathbb{C}^{d_B \times d_B} \text{ unitary.} \tag{12}$$

Then, using $V^\dagger V = \mathbb{1}$,

$$\rho_A = USV^\dagger V S^\dagger U^\dagger = USS^\dagger U^\dagger, \qquad SS^\dagger = \mathrm{diag}(s_1^2, \dots, s_r^2, 0, \dots, 0), \tag{13}$$

and write $S^2 = \mathrm{diag}(s_1^2, \dots, s_r^2)$. Furthermore, since ρ_A is hermitian we can diagonalize it with $\rho_A = ODO^\dagger = US^2U^\dagger$ and $D = \mathrm{diag}(\lambda_1, \dots)$, so that

$$\boxed{\lambda_i = s_i^2}, \tag{14}$$

i.e. the eigenvalues are given by the squared Schmidt values and hence non-negative. Thus, we can conclude together with eq. (11) that

$$\boxed{0 \le \lambda_i \le 1}. \tag{15}$$

1.7 Functions of the density matrix via the spectral theorem

Since ρ_A is hermitian, we can use the spectral theorem to write it as $\rho_A = \sum_i \lambda_i P_i$ where λ_i are its eigenvalues and the eigenvectors $|u_i\rangle$ give $P_i = |u_i\rangle\langle u_i|$, since

$$\rho_A = UDU^\dagger = \big(|u_1\rangle, \dots, |u_n\rangle\big) \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \begin{pmatrix} \langle u_1| \\ \vdots \\ \langle u_n| \end{pmatrix}, \qquad \langle u_i|u_j\rangle = \delta_{ij}. \tag{16}$$

For a function f defined on the spectrum of ρ_A, the spectral functional calculus[1] gives

$$f(\rho_A) \equiv \sum_i f(\lambda_i)\, |u_i\rangle\langle u_i| = U\, \mathrm{diag}\big(f(\lambda_i)\big) U^\dagger. \tag{17}$$

1.8 The von Neumann entanglement entropy

Therefore, for the von Neumann entanglement entropy,

$$\begin{aligned} S(\rho_A) &\equiv -\operatorname{Tr}\big(\rho_A \log \rho_A\big) && (18)\\ &= -\operatorname{Tr}\big(\rho_A\, U \log(\mathrm{diag}(\lambda_i))\, U^\dagger\big) \\ &= -\operatorname{Tr}\big(U\, \mathrm{diag}(\lambda_i)\, \underbrace{U^\dagger U}_{=\mathbb{1}} \log(\mathrm{diag}(\lambda_i)) U^\dagger\big) \\ &= -\sum_i \lambda_i \log \lambda_i = -\sum_i s_i^2 \log s_i^2. && (19) \end{aligned}$$

S is zero for two unentangled systems and non-zero for entangled systems.

[1] See for example Nicholas, J. H. (2008). *Functions of matrices: theory and computation.*

1.9 Example

In our case, for example, $L = 2$, $N_f = 1$, $N_{\rm ph} = 1$:

$$\{|0\rangle, |1\rangle\} \;\&\; \{|01\rangle, |10\rangle\} \quad \text{as bases of } \mathcal{H}_b \;\&\; \mathcal{H}_f, \tag{20}$$

$$\Rightarrow \{|0,01\rangle;\ |0,10\rangle;\ |1,01\rangle;\ |1,10\rangle\} \quad \text{as basis of } \mathcal{H}_b \otimes \mathcal{H}_f, \tag{21}$$

$$\Rightarrow |\psi\rangle = \sum_{i=1}^{2}\sum_{j=1}^{2} \psi_{ij} |i\rangle_b |j\rangle_f \quad \text{with } (\psi)_{ij} = \psi_{ij} \;\&\; \psi \in \mathbb{C}^{2\times 2}, \tag{22}$$

$$\Rightarrow S(\rho_A) = -\sum_{\ell} s_\ell^2 \log s_\ell^2. \tag{23}$$

Note that since $\operatorname{tr}\rho^{AB} = \operatorname{tr}\rho^A = \operatorname{tr}\rho^B = 1$ we must have $\sum_\ell \lambda_\ell = \sum_\ell s_\ell^2 = 1$.

1.10 The Rényi entanglement entropies

The Rényi entropy of order n is defined as

$$S_n(\rho_A) \equiv \frac{1}{1-n}\ln(\operatorname{tr}[\rho_A^n]) \tag{24}$$

$$= \frac{1}{1-n}\ln\left(\sum_{i=1}^{d_A}\lambda_i^n\right), \tag{25}$$

using again, that any hermitian matrix is diagonalizable and the cylic property of the trace.

Claim: The Rényi entropy of order one reduces to the von Neumann entropy.

Proof: We can examine the first order by using l'Hôpital's rule, which yields

$$\begin{aligned} S_1(\rho_A) &= \lim_{n\to 1}\frac{\ln(\operatorname{tr}[\rho_A^n])}{1-n} \overset{\text{H}}{=} \lim_{n\to 1}\frac{\left(\ln\sum_{i=1}^{d_A}\lambda_i^n\right)'}{(1-n)'} \\ &= -\lim_{n\to 1}\frac{\sum_{i=1}^{d_A}\lambda_i^n\ln\lambda_i}{\sum_{i=1}^{d_A}\lambda_i^n} = -\frac{\sum_{i=1}^{d_A}\lambda_i\ln\lambda_i}{\sum_{i=1}^{d_A}\lambda_i} \\ &= -\sum_{i=1}^{d_A}\lambda_i\ln\lambda_i. \end{aligned}$$

■

Claim: $S_n \geq S_m$ for $n < m$.

Proof: Define $z_i(n) = \lambda_i^n / \sum_{j=1}^{d_A}\lambda_j^n$, which is a probability distribution because $0 \leq \lambda_i \leq 1$, and $F(n) = \ln\sum_{j=1}^{d_A}\lambda_j$. We take n to be a continuous variable and differentiate F with

respect to it using again that $\frac{d}{dn}\lambda^n = \lambda^n \ln \lambda$:

$$\begin{aligned} F'(n) &= \frac{\sum_{i=1}^{d_A} \lambda_i^n \ln \lambda_i}{\sum_{j=1}^{d_A} \lambda_j^n} = \sum_i z_i(n) \ln \lambda_i \qquad \text{and} \\ F''(n) &= \sum_i z_i(n) (\ln \lambda_i)^2 - \left(\sum_i z_i(n) \ln \lambda_i \right)^2 \\ &= \mathrm{Var}_{z(n)}(\ln \lambda_i) \geq 0, \end{aligned}$$

which implies, that F' is non-decreasing. Thus, we can follow w.l.o.g for $x \geq a$

$$F(x) - F(n) = \int_n^x F'(s) ds \leq \int_a^x F'(n) ds = F'(n)(x - n),$$

which is equivalent to saying that F lays above all of its tangent, since it is convex. Since $F(1) = \ln \sum_i \lambda_i = \ln 1 = 0$, it follows that

$$F(1) = 0 \leq F(n) + F'(n)(1 - n).$$

We can employ this now to yield

$$\frac{dS(n)}{dn} = \frac{d}{dn} \frac{F(n)}{1-n} = \frac{(1-n)F'(n) + F(n)}{(1-n)^2} \leq 0,$$

which proves, that S_n is non-increasing for $n \neq 1$. Finally, employing l'Hôpital's rule yields for $n = 1$ that

$$\lim_{n \to 1} = -\frac{1}{2} F''(1) \leq 0.$$

■

1.11 References

- M. Nielsen & I. Chuang, *Quantum Computation and Quantum Information* (density operator, partial trace; Box 2.6).
- U. Schollwöck, *The density-matrix renormalization group in the age of matrix product states*, beginning of Chapter 4 (MPS).

2 Computing entanglement entropy with variational quantum states

TODO: add renyi entropy

2.1 Direct computation using the SVD

The entanglement entropy for the ground state can be computed directly as follows:

1. Obtain an approximation of the ground state $|\psi_0\rangle = \sum_\alpha c_\alpha |\alpha\rangle \in \mathcal{H}^{AB} = \mathcal{H}^A \otimes \mathcal{H}^B$ as some parametrized variational state $\psi_\theta(\alpha)$.
2. Use the variational state (e.g. $\psi_\theta(\alpha) = c_\alpha$) to evaluate the amplitudes on every allowed configuration α and normalize the resulting coefficient vector.
3. Reshape the coefficients into the matrix $\psi \in \mathbb{C}^{d_A \times d_B}$, respecting the ordering of the product basis.
4. Compute the SVD of ψ to obtain the Schmidt values s_i, yielding the entropy $S = -\sum_i s_i^2 \log s_i^2$

This approach is also straightforward with exact diagonalization since the ground state is represented by a vector consisting of the desired amplitudes. However, this quickly becomes unfeasible for ED and variational states as the Hilbert space grows exponentially with the system size.

2.2 Constructing the photon reduced density matrix

Given that $d_A \ll d_B$, one can estimate the small reduced density matrix ρ_A and diagonalize it directly. For illustrative purposes, let subsystem A describe a single photon mode and let subsystem B describe a fermionic system.

Let $n = 0, \ldots, N_{\text{ph}}$ denote the photon occupation and σ a fermionic occupation configuration. Define the wavefunction amplitudes by

$$\psi(n, \sigma) \equiv \langle n, \sigma | \psi \rangle \,. \tag{26}$$

For a normalized state, the matrix elements of ρ_A follow from eqs. (9) and (10)

$$(\rho_A)_{nn'} = \sum_\sigma \psi(n, \sigma)\psi^*(n', \sigma) \tag{27}$$

Since variational states do not necessarily provide normalized amplitudes, we might have to normalize them in the above formula. Defining

$$Z \equiv \sum_{m,\sigma} |\psi(m, \sigma)|^2, \tag{28}$$

we can write the normalized reduced density matrix more generally as

$$\boxed{(\rho_A)_{nn'} = \frac{\sum_\sigma \psi(n, \sigma)\psi^*(n', \sigma)}{Z}}. \tag{29}$$

For a normalized state, $Z = 1$.

2.3 A Monte Carlo estimator for the reduced density matrix

For each fermionic configuration σ, define

$$u_n(\sigma) \equiv \frac{\psi(n,\sigma)}{\sqrt{\sum_{m=0}^{N_{\rm ph}} |\psi(m,\sigma)|^2}}, \qquad \boldsymbol{u}(\sigma) = \begin{pmatrix} u_0(\sigma) \\ \vdots \\ u_{N_{\rm ph}}(\sigma) \end{pmatrix} \tag{30}$$

only if the denominator is non-zero. This is almost always the case when using Monte Carlo sampling since these configurations will never be visited, unless you start in one of them. This vector is normalized for every σ with non-zero marginal probability:

$$\boldsymbol{u}(\sigma)^\dagger \boldsymbol{u}(\sigma) = \sum_{n=0}^{N_{\rm ph}} |u_n(\sigma)|^2 = 1. \tag{31}$$

The matrix elements of its outer product are

$$\big(\boldsymbol{u}(\sigma)\boldsymbol{u}(\sigma)^\dagger\big)_{nn'} = \frac{\psi(n,\sigma)\psi^*(n',\sigma)}{\sum_{m=0}^{N_{\rm ph}} |\psi(m,\sigma)|^2}. \tag{32}$$

Now consider the marginal probability distribution of the fermionic configurations,

$$p(\sigma) = \sum_{m=0}^{N_{\rm ph}} p(m,\sigma) \tag{33}$$

$$\equiv \frac{\sum_{m=0}^{N_{\rm ph}} |\psi(m,\sigma)|^2}{\sum_{m,\sigma'} |\psi(m,\sigma')|^2}. \tag{34}$$

Taking the expectation value with respect to this distribution yields

$$\begin{aligned}
\mathbb{E}_{\sigma\sim p(\sigma)}\big[\boldsymbol{u}(\sigma)\boldsymbol{u}(\sigma)^\dagger\big]_{nn'} &= \sum_\sigma \frac{\psi(n,\sigma)\psi^*(n',\sigma)}{\sum_m |\psi(m,\sigma)|^2}\, p(\sigma) \\
&= \sum_\sigma \frac{\psi(n,\sigma)\psi^*(n',\sigma)}{\sum_m |\psi(m,\sigma)|^2}\, \frac{\sum_m |\psi(m,\sigma)|^2}{\sum_{m,\sigma'} |\psi(m,\sigma')|^2} \\
&= \frac{\sum_\sigma \psi(n,\sigma)\psi^*(n',\sigma)}{\sum_{m,\sigma'} |\psi(m,\sigma')|^2} \\
&= (\rho_A)_{nn'}.
\end{aligned} \tag{35}$$

Therefore,

$$\boxed{\rho_A = \mathbb{E}_{\sigma\sim p(\sigma)}\big[\boldsymbol{u}(\sigma)\boldsymbol{u}(\sigma)^\dagger\big]}. \tag{36}$$

Moreover, since we are only interested in an expectation value over σ, we have

$$\begin{aligned}
\mathbb{E}_{(n,\sigma)\sim p(n,\sigma)}\big[F(\sigma)\big] &= \sum_{n,\sigma} p(n,\sigma) F(\sigma) \\
&= \sum_\sigma p(\sigma) F(\sigma) \\
&= \mathbb{E}_{\sigma\sim p(\sigma)}\big[F(\sigma)\big].
\end{aligned} \tag{37}$$

Thus, we can sample either from $p(\sigma)$ or from $p(n,\sigma)$, depending on what is more convenient.

2.4 Computation using variational Monte Carlo

The resulting procedure is:

1. Sample a configuration $(\tilde{n}, \sigma)$ using the VMC sampler, i.e. from the joint distribution

$$p(n, \sigma) = \frac{|\psi(n, \sigma)|^2}{Z}. \tag{38}$$

   Discard $\tilde{n}$ and retain σ. Note, that Z does not need to be known in VMC. The retained configurations follow the marginal distribution $p(\sigma)$ (see eq. (37)).

2. Evaluate all amplitudes $\psi(n, \sigma)$ for $n = 0, \ldots, N_{\text{ph}}$ and construct $\boldsymbol{u}(\sigma)$.

3. Compute $\boldsymbol{u}(\sigma)\boldsymbol{u}(\sigma)^\dagger$. Repeat the preceding steps and average the resulting matrices:

$$\widehat{\rho}_A = \frac{1}{M} \sum_{k=1}^{M} \boldsymbol{u}(\sigma_k)\boldsymbol{u}(\sigma_k)^\dagger \approx \rho_A. \tag{39}$$

   The wide hat indicates that we only obtain an estimate of the true density matrix.

4. Diagonalize $\widehat{\rho}_A$ and use its eigenvalues $\widehat{\lambda}_i$ to estimate the von Neumann entropy:

$$\widehat{S} = -\sum_i \widehat{\lambda}_i \log \widehat{\lambda}_i. \tag{40}$$

2.5 References

I did not use any literature explicitly, except for AI helping me figure out what to do. The following paper employs a similar approach to the one I demonstrate here.

- Tang, Yifan, et al. *Deep quantum Monte Carlo approach for polaritonic chemistry.* The Journal of Chemical Physics 163.3 (2025). (Appendix A.1 and A.3)